Reading Notes: Analytic Combinatorics

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April 27, 2026

Contents

1 COMBINATORIAL STRUCTURES AND ORDINARY GENERATING FUNCTIONS 5

1.1 Symbolic enumeration methods 5
1.2 Admissible constructions and specifications 5
1.3 Integer compositions and partitions 5
1.4 Words and regular languages 14
1.5 Tree structures 14
1.6 Additional constructions 15
1.7 Perspective 15

2 LABELLED STRUCTURES AND EXPONENTIAL GENERATING FUNCTIONS 17

2.1 Labelled classes 17
2.2 Admissible labelled constructions 17
2.3 Surjections, set partitions, and words 17
2.4 Alignments, permutations, and related structures 17
2.5 Labelled trees, mappings, and graphs 17
2.6 Additional constructions 17
2.7 Perspective 17

3 COMBINATORIAL PARAMETERS AND MULTIVARIATE GENERATING FUNCTIONS 19

3.1 An introduction to bivariate generating functions (BGFs) 19
3.2 Bivariate generating functions and probability distributions 19
3.3 Inherited parameters and ordinary MGFs 19
3.4 Inherited parameters and exponential MGFs 19
3.5 Recursive parameters 19
3.6 Complete generating functions and discrete models 19
3.7 Additional constructions 19
3.8 Extremal parameters 19
3.9 Perspective 19

Chapter 1

COMBINATORIAL STRUCTURES AND ORDINARY GENERATING FUNCTIONS

1.1 Symbolic enumeration methods

▷ **I.2.** *Unimodal permutations.*

$$\mathcal{P}^{\mathcal{U}} = \textsc{Set}\left(\mathcal{Z}\right) \star \mathcal{Z}^{\blacksquare} \star \textsc{Set}\left(\mathcal{Z}\right) \tag{1.1}$$

$$P^{U}\left(z\right) = \int_{0}^{z} \left(e^{t} \cdot \frac{d}{dt} t \cdot e^{t}\right) dt = \frac{1}{2}\left(e^{2z} - 1\right) \tag{1.2}$$

which yields

$$P_{n}^{U} = n! \frac{1}{2}\left[z^{n}\right] e^{2z} = n! \frac{1}{2} \frac{2^{n}}{n!} = 2^{n-1} \tag{1.3}$$

Alternatively,

$$P_{n}^{U} = \sum_{k=1}^{n} \binom{n-1}{k-1} = \sum_{j=0}^{n-1} \binom{n-1}{j} = 2^{n-1} \tag{1.4}$$

where k is the position of n in the permutation.

1.2 Admissible constructions and specifications

1.3 Integer compositions and partitions

▷ **I.16.** *Partitions with summands bounded in number and size.* The number of partitions of size n with at most k summands, each at most l, is

$$\left[z^{n}\right] \frac{\left(1-z\right)\left(1-z^{2}\right)\cdots\left(1-z^{k+l}\right)}{\left(\left(1-z\right)\left(1-z^{2}\right)\cdots\left(1-z^{k}\right)\right)\cdot\left(\left(1-z\right)\left(1-z^{2}\right)\cdots\left(1-z^{l}\right)\right)} \tag{1.5}$$

Let $f\left(k,l\right)$ be the generating function in which the coefficient of z^{n} is the number of partitions of size n with at most k summands, each at most l. Consider the largest

possible part in our partition. There are two mutually exclusive cases: the largest part is $< l$, and the largest part is $= l$. This yields the functional equation:

$$f(k,l) = f(k,l-1) + z^l f(k-1,l) \tag{1.6}$$

We can verify that 1.5 satisfies 1.6. Can we derive 1.5 from 1.6? Let $F_k(x)$ be the generating function for $f(k,l)$ with respect to the index l:

$$F_k(x) = \sum_{l=0}^{\infty} f(k,l)\, x^l \tag{1.7}$$

For $k = 0$, there is only one partition of 0 into at most 0 parts (the empty partition). Thus $f(0,l) = 1$ for all $l \geq 0$.

$$F_0(x) = \sum_{l=0}^{\infty} 1 \cdot x^l = \frac{1}{1-x} \tag{1.8}$$

Take 1.6, multiply by x^l, and sum over $l \geq 1$:

$$\sum_{l=1}^{\infty} f(k,l)\, x^l - \sum_{l=1}^{\infty} f(k,l-1)\, x^l = \sum_{l=1}^{\infty} f(k-1,l)\,(zx)^l \tag{1.9}$$

- The first term is $F_k(x) - f(k,0)$. Since $f(k,0) = 1$ (at most 0 size summands), this equals $F_k(x) - 1$.
- The second term is $x \sum_{j=0}^{\infty} f(k,j)\, x^j = xF_k(x)$.
- The right-hand side is $F_{k-1}(zx) - f(k-1,0) = F_{k-1}(zx) - 1$.

Plugging these in:

$$(1-x)\,F_k(x) - 1 = F_{k-1}(zx) - 1 \Longrightarrow F_k(x) = \frac{1}{1-x} F_{k-1}(zx) \tag{1.10}$$

Now we apply this functional relation repeatedly:

$$\begin{aligned} F_k(x) &= \frac{1}{1-x} F_{k-1}(zx) \\ &= \frac{1}{1-x} \cdot \frac{1}{1-zx} F_{k-2}\left(z^2 x\right) \\ &= \frac{1}{(1-x)(1-zx)\cdots(1-z^{k-1}x)} F_0\left(z^k x\right) \end{aligned} \tag{1.11}$$

Since $F_0(x) = \dfrac{1}{1-x}$, we have $F_0\left(z^k x\right) = \dfrac{1}{1-z^k x}$. Thus:

$$F_k(x) = \sum_{l=0}^{\infty} f(k,l)\, x^l = \frac{1}{(1-x)(1-zx)\cdots(1-z^k x)} \tag{1.12}$$

Since all the roots of the denominator are distinct (assuming $|z| < 1$ or that z is a formal variable), we can decompose this into a sum of partial fractions:

$$\frac{1}{\prod_{i=0}^{k}\left(1-z^{i}x\right)}=\sum_{j=0}^{k}\frac{A_{j}}{1-z^{j}x} \tag{1.13}$$

To find a specific A_j, we multiply both sides by $\left(1-z^{j}x\right)$ and then set $x = z^{-j}$:

$$\begin{aligned}A_{j}&=\left[\frac{1-z^{j}x}{\prod_{i=0}^{k}\left(1-z^{i}x\right)}\right]_{x=z^{-j}}\\&=\frac{1}{\prod_{i=0}^{j-1}\left(1-z^{i}z^{-j}\right)\prod_{i=j+1}^{k}\left(1-z^{i}z^{-j}\right)}\\&=\frac{(-1)^{j}\,z^{j(j+1)/2}}{\prod_{i=1}^{j}\left(1-z^{i}\right)\prod_{i=1}^{k-j}\left(1-z^{i}\right)}\end{aligned} \tag{1.14}$$

The coefficient of x^l in $F_k(x)$ is the sum of the coefficients of x^l in each partial fraction. For a term $\dfrac{A_j}{1-z^jx}$, the coefficient is $A_j\left(z^j\right)^l$.

$$\begin{aligned}f(k,l)&=\sum_{j=0}^{k}A_{j}\left(z^{j}\right)^{l}\\&=\sum_{j=0}^{k}\frac{(-1)^{j}\,z^{j(j+1)/2}\cdot z^{jl}}{\prod_{i=1}^{j}\left(1-z^{i}\right)\prod_{i=1}^{k-j}\left(1-z^{i}\right)}\\&=\frac{1}{\prod_{i=1}^{k}\left(1-z^{i}\right)}\sum_{j=0}^{k}\binom{k}{j}_{z}(-1)^{j}\,z^{j(j+1)/2+jl}\end{aligned} \tag{1.15}$$

where $\binom{k}{j}_z$ is the Gaussian binomial coefficient:

$$\begin{aligned}\binom{k}{j}_{z}&=\frac{\prod_{i=1}^{k}\left(1-z^{i}\right)}{\prod_{i=1}^{j}\left(1-z^{i}\right)\prod_{i=1}^{k-j}\left(1-z^{i}\right)}\\&=\frac{\left(1-z^{k}\right)\left(1-z^{k-1}\right)\cdots\left(1-z^{k-j+1}\right)}{\left(1-z\right)\left(1-z^{2}\right)\cdots\left(1-z^{j}\right)}\end{aligned} \tag{1.16}$$

We can rewrite the exponent in the sum:

$$\frac{j\,(j+1)}{2}+jl=\frac{j\,(j-1)}{2}+j\,(l+1) \tag{1.17}$$

Now the sum becomes

$$f(k,l)=\frac{1}{\prod_{i=1}^{k}\left(1-z^{i}\right)}\sum_{j=0}^{k}\binom{k}{j}_{z}(-1)^{j}\,z^{j(j-1)/2}\left(z^{l+1}\right)^{j} \tag{1.18}$$

The q-binomial theorem states that for a variable X:

$$\prod_{i=0}^{k-1}\left(1-z^{i}X\right)=\sum_{j=0}^{k}\binom{k}{j}_{z}(-1)^{j}\,z^{j(j-1)/2}X^{j} \tag{1.19}$$

Plugging $X = z^{l+1}$ into the theorem yields 1.5:

$$\begin{aligned} f(k,l) &= \frac{\left(1-z^{l+1}\right)\left(1-z^{l+2}\right)\cdots\left(1-z^{l+k}\right)}{\left(1-z^{1}\right)\left(1-z^{2}\right)\cdots\left(1-z^{k}\right)} \\ &= \frac{\left(1-z\right)\left(1-z^{2}\right)\cdots\left(1-z^{k+l}\right)}{\left(\left(1-z\right)\left(1-z^{2}\right)\cdots\left(1-z^{k}\right)\right)\cdot\left(\left(1-z\right)\left(1-z^{2}\right)\cdots\left(1-z^{l}\right)\right)} \end{aligned} \tag{1.20}$$

To prove 1.19, let $P_k(X)$ be the product we are expanding:

$$P_k(X) = \prod_{i=0}^{k-1}\left(1-z^{i}X\right) \tag{1.21}$$

We want to show that $P_k(X) = \sum_{j=0}^{k}\binom{k}{j}_z (-1)^j z^{j(j-1)/2}X^j$. Consider the relationship between $P_k(X)$ and $P_{k-1}(X)$. By definition:

$$P_k(X) = \left(1-z^{k-1}X\right)P_{k-1}(X) \tag{1.22}$$

Now notice what happens if we replace X with zX in $P_k(X)$:

$$P_k(zX) = \prod_{i=0}^{k-1}\left(1-z^{i+1}X\right) = \frac{1-z^kX}{1-X}P_k(X) \tag{1.23}$$

Rearranging this gives us a functional equation for $P_k(X)$:

$$(1-X)P_k(zX) = \left(1-z^kX\right)P_k(X) \tag{1.24}$$

Let $P_k(X) = \sum_{j=0}^{k} c_jX^j$. We substitute this power series into the functional equation:

$$(1-X)\sum_{j=0}^{k} c_j(zX)^j = \left(1-z^kX\right)\sum_{j=0}^{k} c_jX^j \tag{1.25}$$

Now, we compare the coefficients of X^j on both sides. For a specific power j:

$$c_jz^j - c_{j-1}z^{j-1} = c_j - c_{j-1}z^k \tag{1.26}$$

Rearranging to solve for c_j in terms of c_{j-1}:

$$c_j = c_{j-1}\frac{z^{j-1}-z^k}{z^j-1} = -c_{j-1}z^{j-1}\frac{1-z^{k-j+1}}{1-z^j} \tag{1.27}$$

Starting from the base case $c_0 = P_k(0) = 1$, we can find c_j by iteration:

$$c_1 = -c_0z^0\frac{1-z^k}{1-z^1}, \quad c_2 = -c_1z^1\frac{1-z^{k-1}}{1-z^2} = (-1)^2z^{0+1}\frac{\left(1-z^k\right)\left(1-z^{k-1}\right)}{\left(1-z^1\right)\left(1-z^2\right)} \tag{1.28}$$

The exponent of z becomes $0+1+2+\cdots+(j-1)=\dfrac{j(j-1)}{2}$. The fraction is the definition of the Gaussian binomial coefficient $\binom{k}{j}_z$. Thus, the coefficient is:

$$c_j=(-1)^j\, z^{j(j-1)/2}\binom{k}{j}_z \tag{1.29}$$

Plugging the coefficients back into the sum, we obtain:

$$\prod_{i=0}^{k-1}\left(1-z^iX\right)=\sum_{j=0}^{k}\binom{k}{j}_z(-1)^j\, z^{j(j-1)/2}X^j \tag{1.30}$$

This completes the proof.

▷ **I.17.** *Systems of linear Diophantine inequalities.*

$$\{x_1\geq 0,\ x_2\geq 2x_1,\ x_3\geq 2x_2,\ x_4\geq 2x_3\} \tag{1.31}$$

We introduce non-negative slack variables $y_j\in\mathbb{Z}_{\geq 0}$ that measure the excess of each term over its minimum bound:

$$x_1=y_1 \tag{1.32}$$
$$x_2=2x_1+y_2=2y_1+y_2 \tag{1.33}$$
$$x_3=2x_2+y_3=4y_1+2y_2+y_3 \tag{1.34}$$
$$x_4=2x_3+y_4=8y_1+4y_2+2y_3+y_4 \tag{1.35}$$

The size of the composition is $S=x_1+x_2+x_3+x_4=15y_1+7y_2+3y_3+y_4$. Since the variables y_1,y_2,y_3,y_4 are independent and each ranges freely over the non-negative integers, the OGF for the class $\mathcal{F}$ is the product of the generating functions for the individual y_j:

- For $15y_1$: $\displaystyle\sum_{y_1=0}^{\infty}\left(z^{15}\right)^{y_1}=\frac{1}{1-z^{15}}$
- For $7y_2$: $\displaystyle\sum_{y_2=0}^{\infty}\left(z^{7}\right)^{y_2}=\frac{1}{1-z^{7}}$
- For $3y_3$: $\displaystyle\sum_{y_3=0}^{\infty}\left(z^{3}\right)^{y_3}=\frac{1}{1-z^{3}}$
- For y_4 : $\displaystyle\sum_{y_4=0}^{\infty}\left(z^{1}\right)^{y_4}=\frac{1}{1-z^{1}}$

Multiplying these together yields the desired OGF:

$$F(z)=\frac{1}{(1-z)(1-z^3)(1-z^7)(1-z^{15})} \tag{1.36}$$

$$\{x_1 + x_2 \leq x_3\} \tag{1.37}$$

Following the same approach as in the previous problem, we use slack variables to convert the inequality into an equality of independent parts. The constraint $x_3 \geq x_1 + x_2$ can be rewritten by introducing a non-negative integer $y \geq 0$:

$$x_3 = x_1 + x_2 + y \tag{1.38}$$

The size of the composition is $S = x_1 + x_2 + x_3 + x_4 = 2x_1 + 2x_2 + y + x_4$. Since the variables x_1, x_2, y, x_4 are independent and each ranges freely over the non-negative integers, the OGF is the product of the generating functions for the individual terms in the sum:

- For $2x_1 : \dfrac{1}{1-z^2}$
- For $2x_2 : \dfrac{1}{1-z^2}$
- For $y : \dfrac{1}{1-z}$
- For $x_4 : \dfrac{1}{1-z}$

Multiplying these together gives:

$$F(z) = \frac{1}{(1-z^2)^2 (1-z)^2} \tag{1.39}$$

$$\{x_1 + x_2 \geq x_3\} \tag{1.40}$$

Because the inequality does not immediately suggest a single slack variable to absorb the excess, the most robust approach is to use the complement method. We know the OGF for four unrestricted variables (x_1, x_2, x_3, x_4) with $x_i \geq 0$ is:

$$U(z) = \frac{1}{(1-z)^4} \tag{1.41}$$

The set of all such compositions partitions naturally into two disjoint subsets: those satisfying $x_1 + x_2 < x_3$ (i.e., $x_1 + x_2 + 1 \leq x_3$), and those satisfying $x_1 + x_2 \geq x_3$ (our target class). We find the OGF for the complement $x_1 + x_2 < x_3$. Set $x_3 = x_1 + x_2 + 1 + y$, where $y \geq 0$. The total sum S is:

$$S = x_1 + x_2 + (x_1 + x_2 + 1 + y) + x_4 = 2x_1 + 2x_2 + y + x_4 + 1 \tag{1.42}$$

The OGF for this complement, which we denote by $C(z)$, is:

$$C(z) = \frac{z}{(1-z^2)^2 (1-z)^2} \tag{1.43}$$

The OGF for our constraint is then:

$$F(z) = U(z) - C(z) = \frac{1}{(1-z)^4} - \frac{z}{(1-z^2)^2 (1-z)^2} = \frac{1+z+z^2}{(1-z)^4 (1+z)^2} \tag{1.44}$$

$$\{x_1 + x_2 \leq x_3 + x_4\} \tag{1.45}$$

To find the OGF for $\{x_1 + x_2 \leq x_3 + x_4\}$, we can use the complement method combined with the symmetry of the variables. Let $A = x_1 + x_2$ and $B = x_3 + x_4$. We seek the OGF for compositions satisfying $A \leq B$. In the universe of all possible compositions (x_1, x_2, x_3, x_4), there are three mutually exclusive cases: 1. $A < B$, 2. $A > B$, 3. $A = B$. By symmetry, the OGF for case 1 must be identical to the OGF for case 2, since A and B are built from the same number of variables with the same weights. Denote this common OGF by $G_<(z)$, and let $G_=(z)$ be the OGF for case 3. The total unrestricted OGF is $U(z) = \dfrac{1}{(1-z)^4}$. Therefore:

$$U(z) = 2G_<(z) + G_=(z) \tag{1.46}$$

Our target OGF is $F(z) = G_<(z) + G_=(z)$. Solving the system gives:

$$F(z) = \frac{U(z) + G_=(z)}{2} \tag{1.47}$$

Now $G_=(z)$ is the OGF for compositions in which the sum of the first two variables equals the sum of the last two.

- The number of compositions with $x_1 + x_2 = n$ is $[z^n]\dfrac{1}{(1-z)^2} = n + 1$.
- The number of compositions with $x_3 + x_4 = n$ is the same.
- For the equality $x_1 + x_2 = x_3 + x_4$ to hold, both sums must equal some $n \geq 0$.

$$G_=(z) = \sum_{n=0}^{\infty} (n+1)^2 z^{2n} = \frac{1+z^2}{(1-z^2)^3} \tag{1.48}$$

Substituting $U(z)$ and $G_=(z)$ into our formula yields:

$$F(z) = \frac{1}{2}\left[\frac{1}{(1-z)^4} + \frac{1+z^2}{(1-z^2)^3}\right] = \frac{1+z+2z^2}{(1-z)^4(1+z)^3} \tag{1.49}$$

$$\{x_1 \leq x_2,\ x_2 \geq x_3,\ x_3 \leq x_4\} \tag{1.50}$$

Because the inequalities alternate direction (up, down, up), we cannot use a straightforward chain of slack variable substitutions as we did for $x_1 \leq x_2 \leq x_3 \leq x_4$. Instead, we apply the inclusion–exclusion principle to the constraints. We begin with the universe of all unrestricted compositions, whose OGF is $U(z) = \dfrac{1}{(1-z)^4}$. We wish to enforce three conditions:

- $C_1 : x_1 \leq x_2$
- $C_2 : x_2 \geq x_3$
- $C_3 : x_3 \leq x_4$

However, it is simpler to work with chains of "less than or equal" inequalities. Consider the complement of the downward middle step. Take the set of all compositions satisfying both $x_1 \leq x_2$ and $x_3 \leq x_4$. From this set, we subtract those compositions that violate the middle constraint (i.e., $x_2 < x_3$).
The OGF for the set defined by $x_1 \leq x_2$ and $x_3 \leq x_4$: These two pairs are independent.

- For $x_1 \leq x_2$, set $x_2 = x_1 + y_1$. The total contribution is $2x_1 + y_1$, so the OGF is $\frac{1}{(1-z^2)(1-z)}$.
- For $x_3 \leq x_4$, the OGF is identically $\frac{1}{(1-z^2)(1-z)}$.

Since the two pairs are independent, we multiply their OGFs:

$$A(z) = \frac{1}{(1-z^2)^2(1-z)^2} \tag{1.51}$$

Now we find the OGF for the violation $x_1 \leq x_2 < x_3 \leq x_4$ by introducing slack variables:

- $x_1 = y_1$
- $x_2 = y_1 + y_2$
- $x_3 = y_1 + y_2 + 1 + y_3$
- $x_4 = y_1 + y_2 + 1 + y_3 + y_4$

Summing the four expressions gives:

$$S = 4y_1 + 3y_2 + 2y_3 + y_4 + 2 \tag{1.52}$$

The OGF for this violation is therefore:

$$V(z) = \frac{z^2}{(1-z^4)(1-z^3)(1-z^2)(1-z)} \tag{1.53}$$

The OGF for the zigzag constraint is the OGF for the two independent pairs minus the OGF for the cases where the peak at x_2 fails:

$$\begin{aligned} F(z) &= \frac{1}{(1-z^2)^2(1-z)^2} - \frac{z^2}{(1-z^4)(1-z^3)(1-z^2)(1-z)} \\ &= \frac{1}{(1-z)(1-z^2)}\left[\frac{1}{(1-z)(1-z^2)} - \frac{z^2}{(1-z^3)(1-z^4)}\right] \end{aligned} \tag{1.54}$$

▷ **I.18.** *Odd versus distinct summands.* We establish a one-to-one correspondence between partitions into odd parts and partitions into distinct parts.

- Odd → Distinct
  1. Group identical odd parts together.
  2. If an odd part k appears m times, write m in binary: $m = 2^{a_1} + 2^{a_2} + \cdots$ with distinct powers of two.

3. Replace the m copies of k by $2^{a_1} \cdot k,\ 2^{a_2} \cdot k,\ \cdots$

- Distinct $\rightarrow$ Odd

1. Factor each part as $2^a \times$ odd.
2. Split it into 2^a copies of that odd factor.
3. Do this for all parts; now all parts are odd.
4. Sort the resulting parts to obtain a partition into odd parts.

As an example:

$$\begin{aligned} 3+3+3+3+3+1+1 &= 5 \times 3 + 2 \times 1 \\ &= \left(2^2 + 2^0\right) \times 3 + 2^1 \times 1 \\ &= 4 \times 3 + 1 \times 3 + 2 \times 1 \\ &= 12 + 3 + 2 \end{aligned} \tag{1.55}$$

$$\begin{aligned} 12+3+2 &= 4 \times 3 + 3 + 2 \times 1 \\ &= (3+3+3+3) + 3 + (1+1) \\ &= 3+3+3+3+3+1+1 \end{aligned} \tag{1.56}$$

▷ **I.19.** *Euler's pentagonal number theorem.* To find a recurrence for the partition numbers, we consider the reciprocal of the generating function $P(z)$:

$$A(z) = \prod_{n=1}^{\infty} (1 - z^n) \tag{1.57}$$

Euler's Pentagonal Number Theorem states that this product expands into the following sparse power series:

$$\begin{aligned} A(z) &= \left(\sum_{k=-\infty}^{\infty} (-1)^k z^{k(3k+1)/2} \right) \\ &= \left(\sum_{k=-\infty}^{\infty} (-1)^k z^{k(3k-1)/2} \right) \\ &= 1 - z^1 - z^2 + z^5 + z^7 - z^{12} - z^{15} + \cdots \end{aligned} \tag{1.58}$$

The exponents $g_k = \dfrac{k(3k-1)}{2}$ are the generalized pentagonal numbers. The product of $P(z)$ and $A(z)$ must equal 1:

$$\left(\sum_{n=0}^{\infty} p(n) z^n \right) \left(\sum_{k=-\infty}^{\infty} (-1)^k z^{g_k} \right) = 1 \tag{1.59}$$

Comparing the coefficient of z^n on both sides yields the algorithmically useful recurrence:

$$p(n) = p(n-1) + p(n-2) - p(n-5) - p(n-7) + p(n-12) + p(n-15) \cdots \tag{1.60}$$

A Python implementation is provided below:

```python
def compute_partitions(N):
P = [0] * (N + 1)
P[0] = 1  # Base case: p(0) = 1

for i in range(1, N + 1):
    k = 1
    while True:
        # Calculate pentagonal numbers for k and -k
        g_k_pos = k * (3 * k - 1) // 2
        g_k_neg = k * (3 * k + 1) // 2

        # Determine sign: +1 for k=1,2 (terms 1,2), -1 for k=3,4 (terms 3,4)...
        # Simplest way: (-1)^(k-1)
        sign = 1 if k % 2 == 1 else -1

        # Add p(i - g_k_pos)
        if g_k_pos <= i:
            P[i] += sign * P[i - g_k_pos]
        else:
            break # No more terms will fit

        # Add p(i - g_k_neg)
        if g_k_neg <= i:
            P[i] += sign * P[i - g_k_neg]
        else:
            break

        k += 1
return P
```

1.4 Words and regular languages

1.5 Tree structures

▷ **I.42.** *Faster determination of Schröder numbers.*

$$S(z) = \frac{1}{4}\left(1 + z - \sqrt{1 - 6z + z^2}\right) \tag{1.61}$$

Consider in general

$$p(z)\,S(z) + q(z) = \sqrt[k]{r(z)} \tag{1.62}$$

where $p(z)$, $q(z)$, and $r(z)$ are functions of z. Taking the derivative with respect to z,

$$p'S + pS' + q' = \frac{1}{k} r^{\frac{1}{k}-1} r' \tag{1.63}$$

Multiplying both sides by $r(z)$,

$$p'rS + prS' + q'r = \frac{1}{k} r^{\frac{1}{k}} r' = \frac{1}{k}(pS + q)\,r' \tag{1.64}$$

Rearranging,

$$prS' + \left(p'r - \frac{1}{k}pr'\right) S = \frac{1}{k}qr' - q'r \tag{1.65}$$

Plugging $p(z) = -4$, $q(z) = 1 + z$, $k = 2$, and $r(z) = 1 - 6z + z^2$ into the equation and simplifying yields

$$\left(z^2 - 6z + 1\right) S' + (3 - z) S = 1 - z \tag{1.66}$$

from which the recurrence follows.

1.6 Additional constructions

1.7 Perspective

Chapter 2

LABELLED STRUCTURES AND EXPONENTIAL GENERATING FUNCTIONS

2.1 Labelled classes

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Chapter 3

COMBINATORIAL PARAMETERS AND MULTIVARIATE GENERATING FUNCTIONS

Content for chapter 3 goes here...

3.1 An introduction to bivariate generating functions (BGFs)

3.2 Bivariate generating functions and probability distributions

3.3 Inherited parameters and ordinary MGFs

3.4 Inherited parameters and exponential MGFs

3.5 Recursive parameters

3.6 Complete generating functions and discrete models

3.7 Additional constructions

3.8 Extremal parameters

3.9 Perspective